2 **(a)** Define the moment of a force about a point.

.....

..... [1]

(b) A tree of mass 270 kg grows out of sloping ground and is supported by a post, as shown in Figure 2.1.

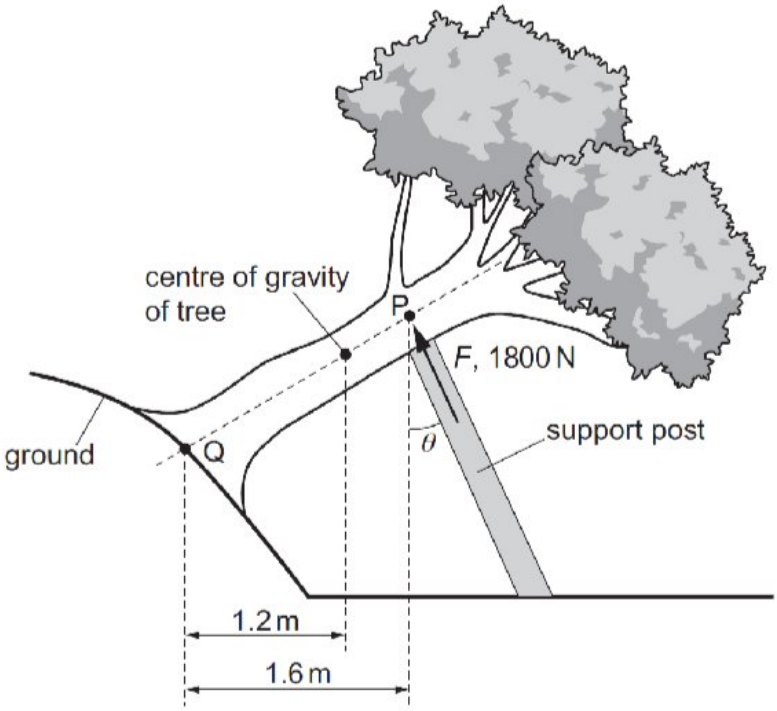


Figure 2.1 (not to scale)

The ground applies a total force *R* on the tree at point Q. The centre of gravity of the tree is a horizontal distance of 1.2 m from Q.

The post applies a force *F* of 1800 N perpendicular to the line PQ. The line of action of *F* passes through point P at an angle *θ* to the vertical. P is a horizontal distance of 1.6 m from Q.

The tree is in equilibrium and all forces act on the tree in the same plane.

(i) Calculate angle *θ*.

θ = ...° [3]

(ii) State and explain whether the total force *R* at point Q is horizontal.

.....

.....

..... [2]

[Total: 6]